



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to A1 FUELS INDIAN OIL DEAL

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

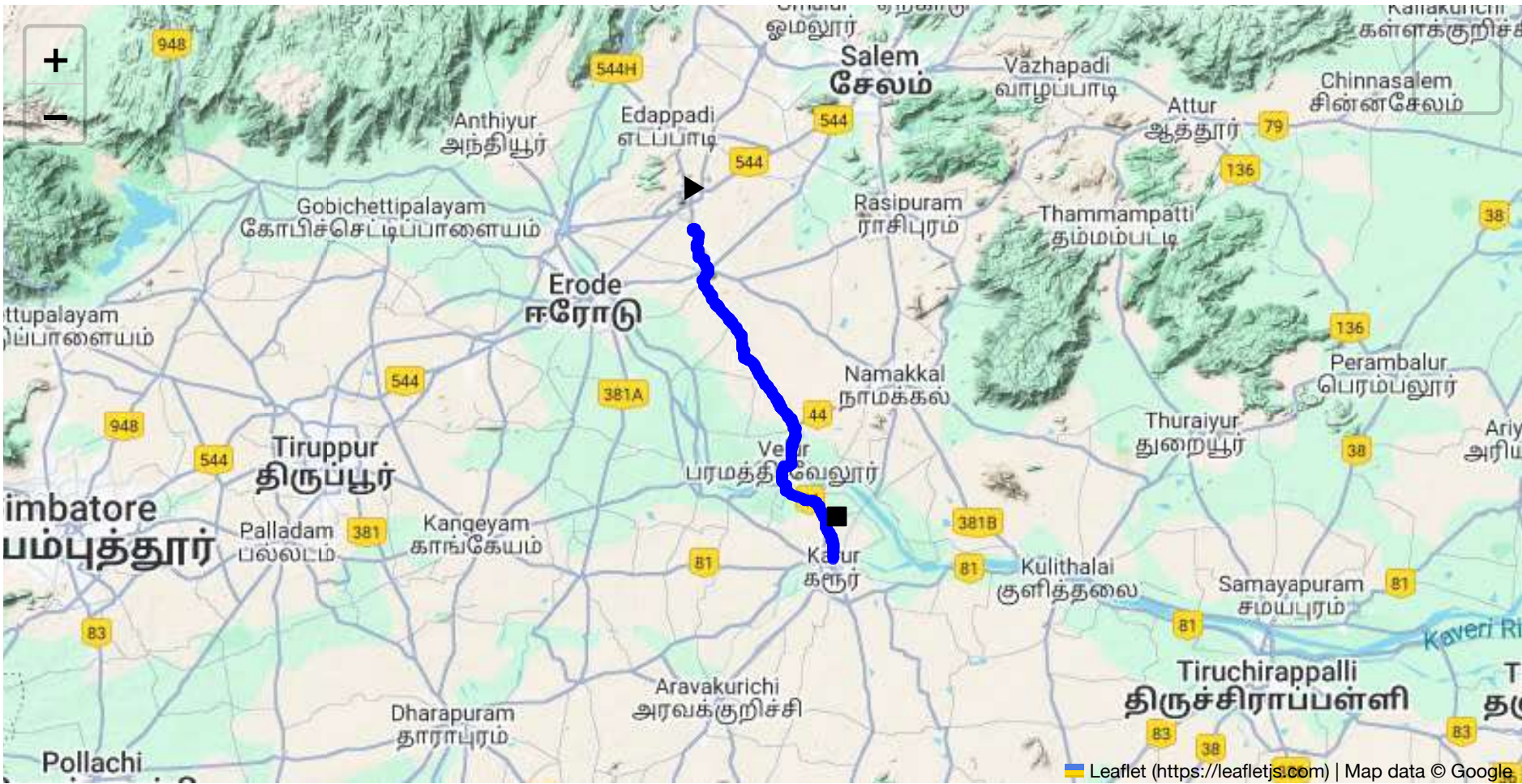
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 63.57 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours

Start: (11.4381, 77.8734)
End: (10.971734, 78.078549)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 3, Salem Main Rd, Kamadenu Nagar, passes predominantly through NH544, eventually joining NH44 to reach Karur. This route is a major transport channel with relatively high traffic flow, providing connections between industrial and urban areas in Tamil Nadu.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Typical Weather:** The region generally experiences a tropical climate, with hot summers, moderate winters, and substantial rainfall during the monsoon season.
- **Potential Hazards:**
 - **Monsoon Season (June to September):** Heavy rainfall may cause waterlogging and reduced visibility.
 - **Summer (April to June):** High temperatures might lead to road surface softening, affecting tire traction.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically between 8-10 AM and 5-8 PM during weekdays.
- **Congestion-Prone Areas:**
 - Sangagiri junctions
 - Salem Main Road toward Karur is known for industrial traffic, which may slow movement.

4. Assessment of Road Quality and Infrastructure

- **NH544 & NH44:** Generally well-maintained but be cautious of:

- Occasional potholes, especially post-monsoon.
- Construction zones that periodically arise, causing temporary detours.

5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route Via SH79 and SH21:** This route might be used during emergencies involving road blockages or heavy congestion.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permit Requirements:** Ensure that all permits for heavy vehicles carrying hazardous materials are up-to-date.
- **Specific Routes:** Certain routes within towns may be restricted for hazardous material transport during peak times or specific conditions.

7. Overview of Historical Incidents

- Reported incidents in the past involve:
 - Overloads leading to tire bursts.
 - Occasional spillage incidents due to improper load securing.

8. Environmental Considerations and Sensitive Areas

- **Forested Zones:** Some segments along NH544 might pass near green cover.
- **Water Bodies:** Ensure no spillage into water bodies along the route, particularly during the monsoon.

9. Analysis of Communication Coverage

- **Potential Dead Zones:** Remote areas might have weak network signals. Equip drivers with offline maps and emergency communication tools.

10. Estimated Emergency Response Times

- **Sangagiri Township and NH544:** Around 20-30 minutes depending on the traffic.
- **Remote Segments of NH44:** Could take up to 45 minutes to an hour for emergency services arrival.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Karur is generally considered safe for transporting heavy vehicles and hazardous materials, with reasonable infrastructure and emergency support. However, drivers should remain vigilant about:

- Weather conditions and environmental factors during the monsoon.
- Anticipated congestion during peak hours.
- Ensure compliance with local transport regulations for hazardous materials. Overall, preparation, careful monitoring of communication equipment, and route updates are critical to maintaining safety across the journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16755, 78.01568	30 KM/Hr
7	Turn	High	11.14545, 78.02208	15 KM/Hr
8	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
19	school	Cheran Matriculation Higher Secondary School	10.9971306, 78.0743488	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16755, 78.01568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.14480, 78.02291

